

Model Predictive Control for a Rocket Booster Landing

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Project Objective

- Design, build, and test an optimization-based controller for reusable rocket booster landing
 - Model system dynamics with gravity, drag, variable mass, and thrust constraints
 - Implementing a Model Predictive Controller (MPC) and a PD baseline in closed loop simulation
 - Evaluate both controllers on landing target accuracy, robustness to wind disturbances, fuel efficiency



Challenges?

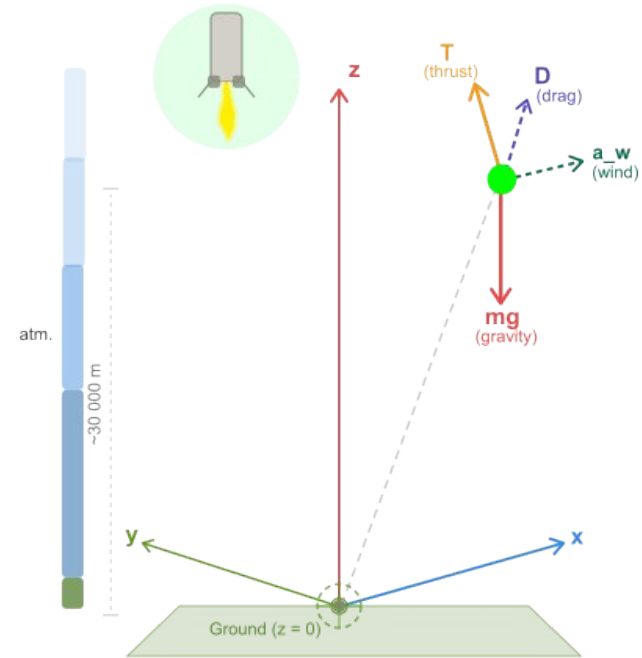
- SLOW DOWN!!!
- Navigate to target
- Don't run out of fuel

Project Setup

Assumptions:

- Point mass model
- Exponential atmosphere/Simple drag
- Wind modeled as constant lateral acceleration

Parameter	Symbol	Value
Initial Mass	m_0	70,000 kg
Dry Mass	m_{dry}	22,000 kg
Specific Impulse	I_{sp}	280 s
Gravity	g	9.81 m/s ²
Atmospheric Density	ρ_0	1.225 kg/m ³
Scale height	H	8500 m
Drag area parameter	$C_D A$	10 m ²



Initial Conditions

Altitude: 30,000 m

Horizontal Offset: $\sim 2,750$ m from target

Velocity: ~ 250 m/s downward (small lateral components)

System Model

State Vector & Control Input

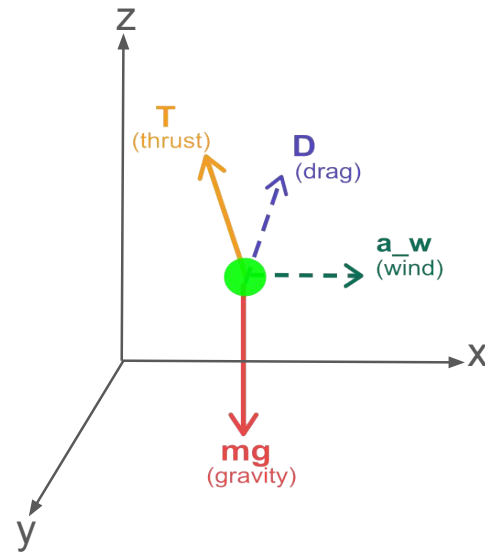
$$x = [r_x \quad r_y \quad r_z \quad v_x \quad v_y \quad v_z \quad m]^T \quad T = [T_x \quad T_y \quad T_z]^T$$

Equations of Motion: Euler Propagation

$$\dot{r} = v \quad \dot{v} = \frac{T}{m} + \begin{bmatrix} 0 \\ 0 \\ -g \end{bmatrix} + a_D + a_w \quad \dot{m} = -\frac{\|T\|}{I_{sp}g_0}$$

Atmosphere and Drag Model

$$\rho(r_z) = \rho_0 e^{-r_z/H} \quad a_D = -\frac{1}{2m} \rho C_D A \|v\| v$$



Constraints, Landing Objectives

- Drive the booster to the target position and velocity with minimum fuel expenditure, subject to physical constraints.



Thrust - 2,535,000 N Maximum (845kN * 3)

$$0 \leq \|T_k\| \leq T_{\max}$$

Thrust tilt - 20° Maximum

$$\theta_k \leq \theta_{\max}$$

Final Mass - 22,000 kg Minimum

$$m_k \geq m_{\text{dry}}$$

Altitude $r_z \geq 0$

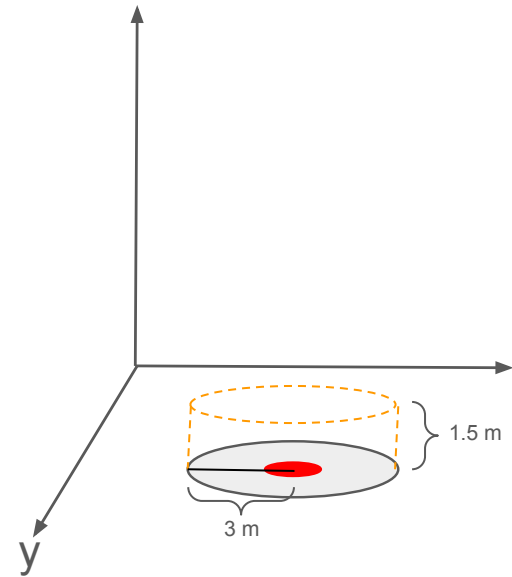
$$r_{z,k} \geq 0$$

Engine Cutoff Condition

$$\sqrt{r_x^2 + r_y^2} \leq 3 \text{ m},$$

$$r_z \leq 1.5 \text{ m},$$

$$\|v\| \leq 1 \text{ m/s}.$$



PD controller

Baseline - A proportional-derivative controller tracks a pre-planned reference trajectory from the initial position to the landing target.

Desired Acceleration

$$a_{\text{cmd}} = K_p(r_{\text{ref}} - r) + K_d(v_{\text{ref}} - v) + \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix}$$

Desired Thrust

$$T = ma_{\text{cmd}}$$

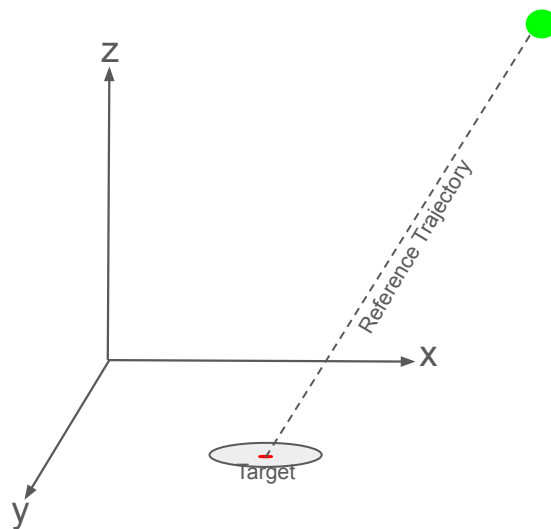
Difference from MPC

No lookahead

- Reacts to current error only

No explicit fuel optimization

- Thrust is whatever the PD law demands



* Same thrust magnitude, engine tilt, and fuel limits from the MPC are applied to the PD controller.

MPC Modeling

At every time step, the MPC solves a constrained finite-horizon optimization problem, applies the first command, then replans from the new state.

Initial State

$$x(0) = [2500 \quad -1200 \quad 30000 \quad -20 \quad 10 \quad -250 \quad 70000]^T$$

Target State

$$r_f = [0 \quad 0 \quad 0]^T, \quad v_f = [0 \quad 0 \quad 0]^T$$

Error Vector

$$e_k = [r_{x,k} \quad r_{y,k} \quad r_{z,k} \quad v_{x,k} \quad v_{y,k} \quad v_{z,k}]^T$$

Time Horizon - 20 seconds

$$\Delta t = 1 \text{ s.}$$

MPC Cost

$$J = \sum_{k=0}^{N-1} (e_k^T Q e_k + R_{\text{cmd}} \|c_k\|^2 + R_{\text{fuel}} \rho_k^2) \Delta t + e_N^T P e_N$$

Running Cost Penalizes

- Position & Velocity Error
- Command Input
- Fuel Use

Terminal Cost Large Penalty

- Final Position
- Velocity Error

Optimizer Command

$$c_k = [\rho_k \quad p_{x,k} \quad p_{y,k}]^T$$

ρ_k : Controls thrust magnitude

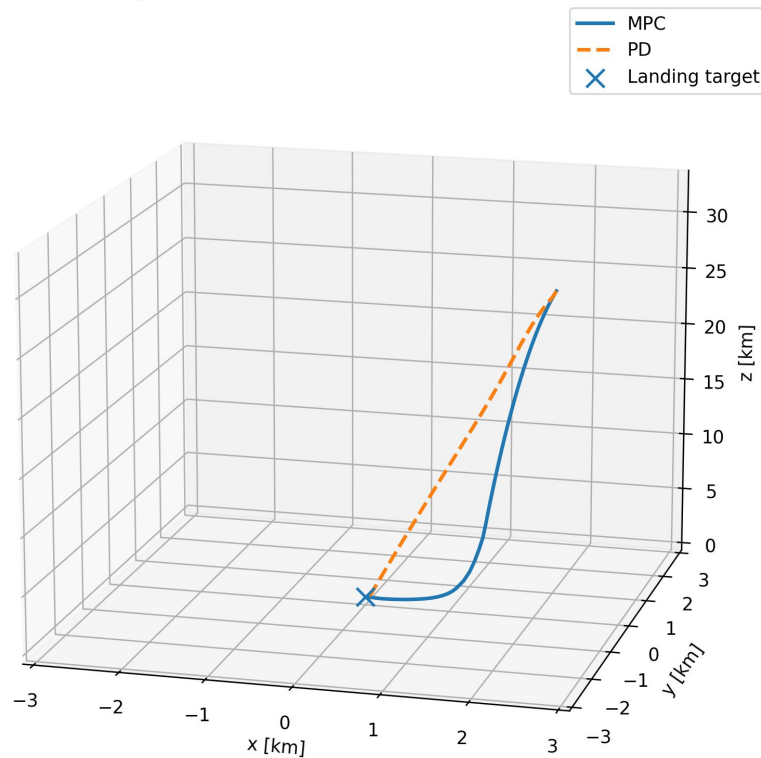
$p_{x,k}, p_{y,k}$: Control thrust direction

Nominal (ideal) results

$$x(0) = [2500 \quad -1200 \quad 30000 \quad -20 \quad 10 \quad -250 \quad 70000]^T$$

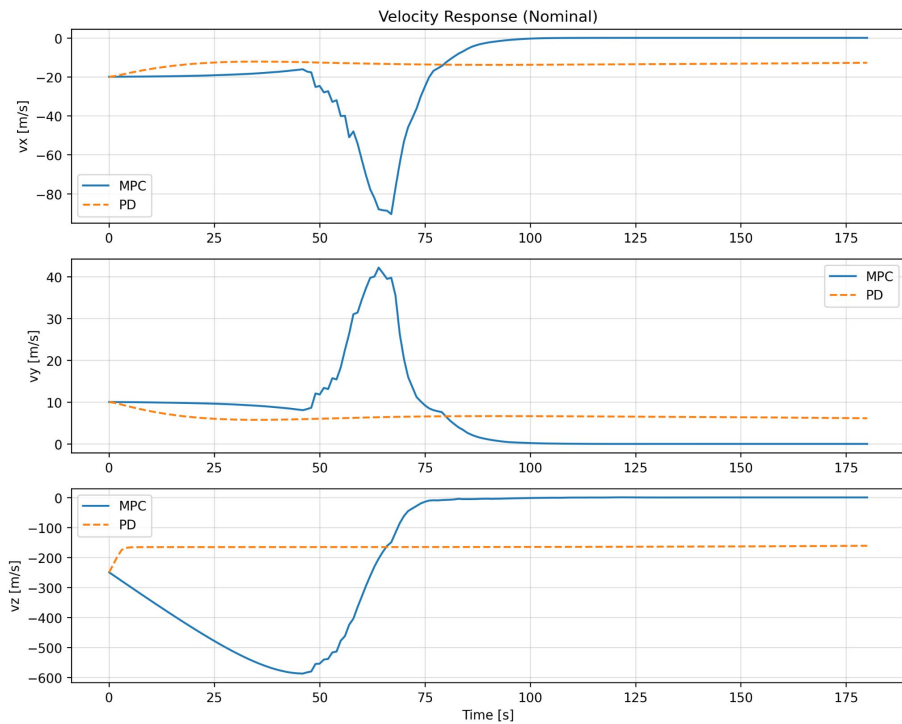
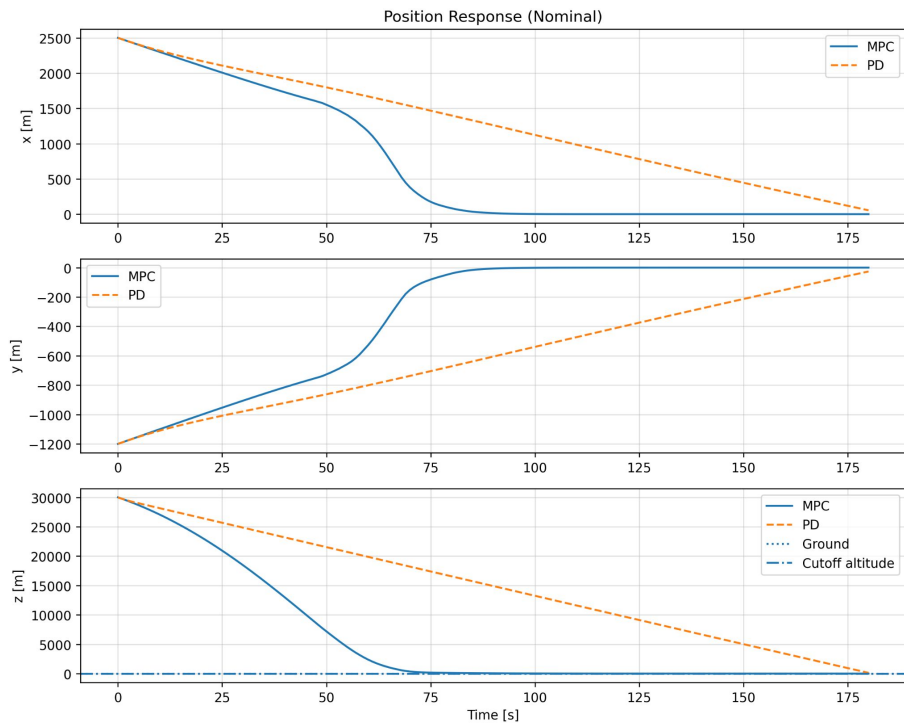
Nominal (ideal) results

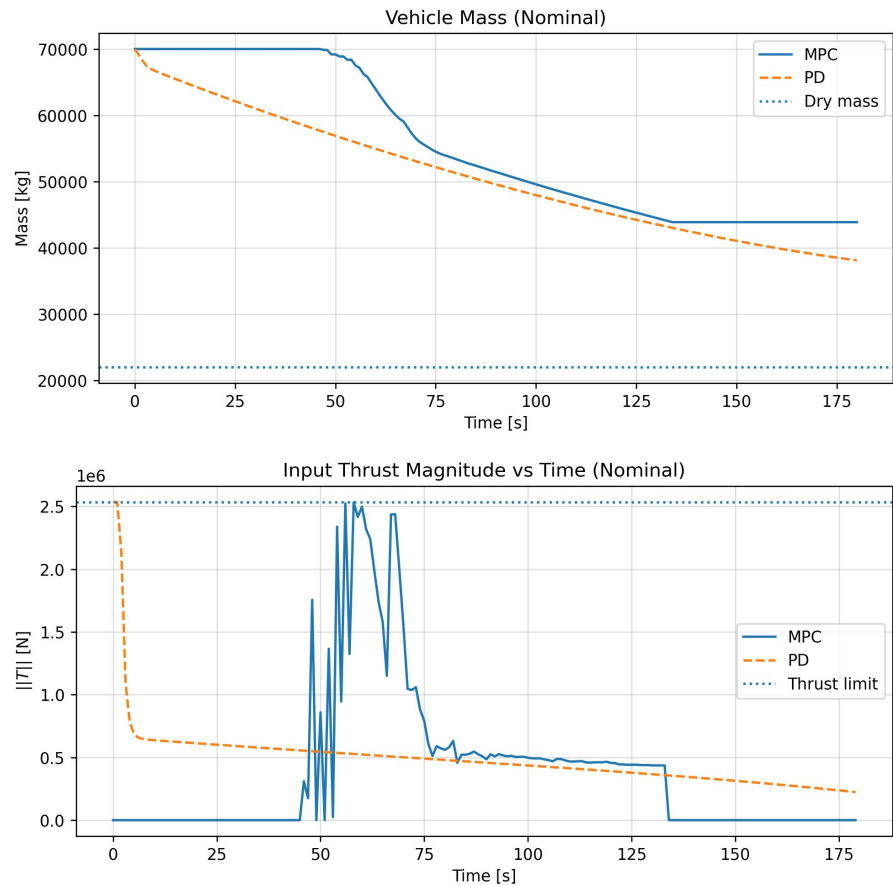
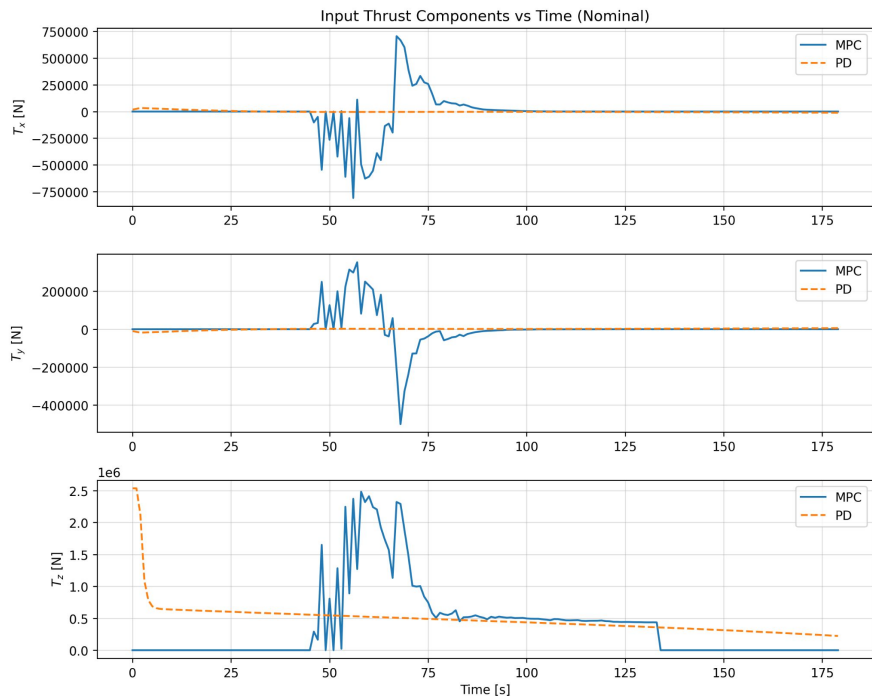
$x(0) = [2500 \quad -1200 \quad 30000 \quad -20 \quad 10 \quad -250 \quad 70000]^T$ 3D Landing Trajectory (Nominal)



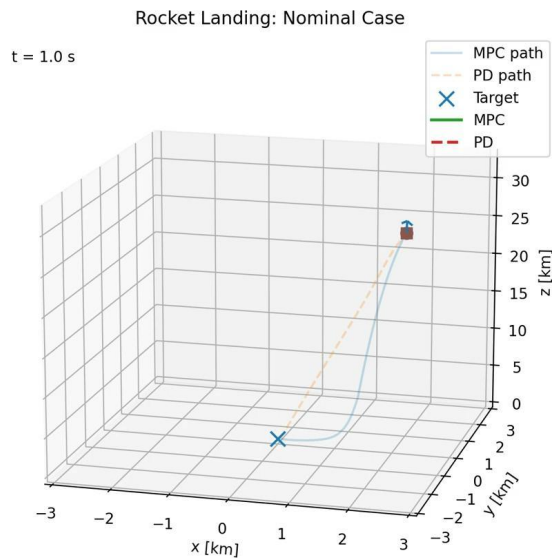
Nominal (ideal) results

	MPC	PD
Final Position Error [m]	1.453	148.734
Final Horizontal Error [m]	0.004	60.532
Final Altitude [m]	1.453	135.860
Final Speed [m/s]	0.122	162.090
Fuel Used [kg]	26124.075	31891.998
Fuel Remaining [kg]	21875.925	16108.002
Max Thrust [N]	2532735.227	2535000.000
Max Tilt Angle [deg]	20.000	3.280

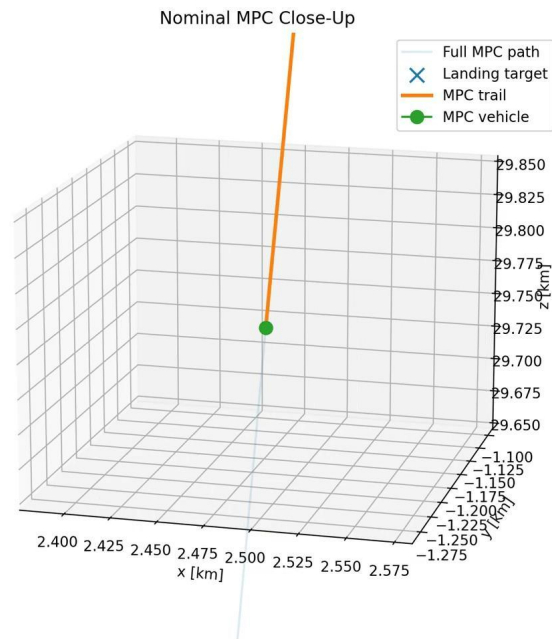




Nominal Animation(s)



t = 1.0 s
x = 2480.0 m
y = -1190.0 m
z = 29750.0 m
speed = 260.61 m/s
|T| = 0.0 N



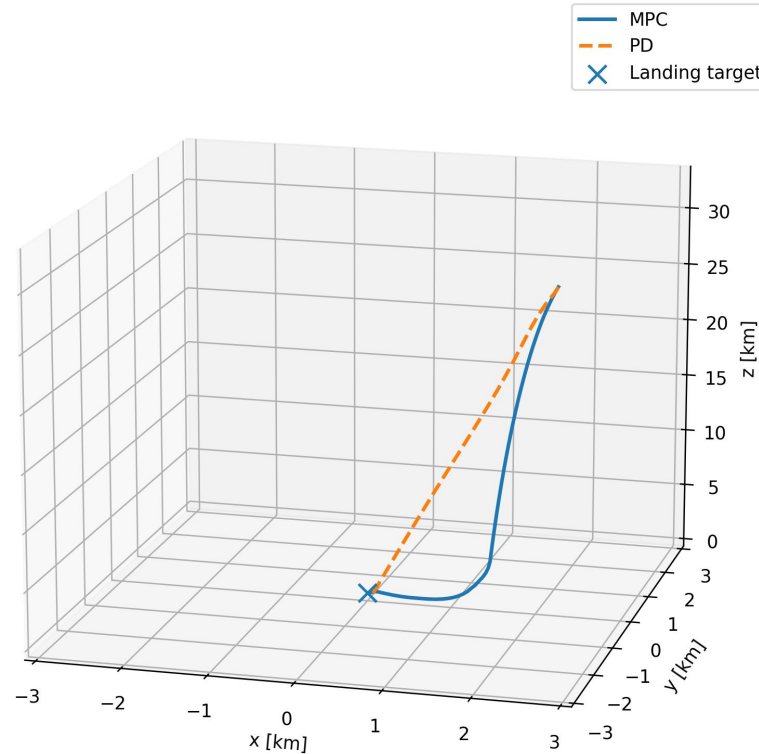
Robustness Study - Wind disturbances

$$a_w = [0.10 \quad -0.05 \quad 0]^T \text{ m/s}^2.$$

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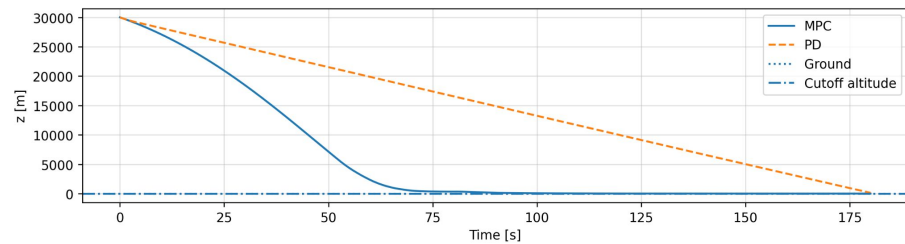
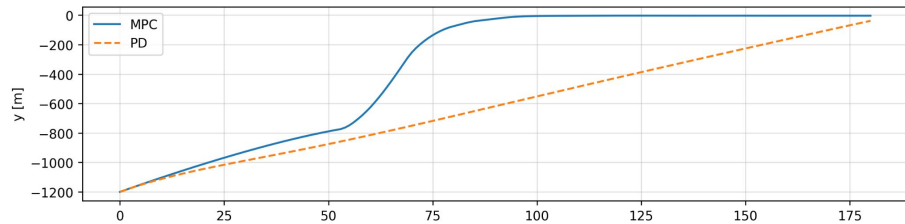
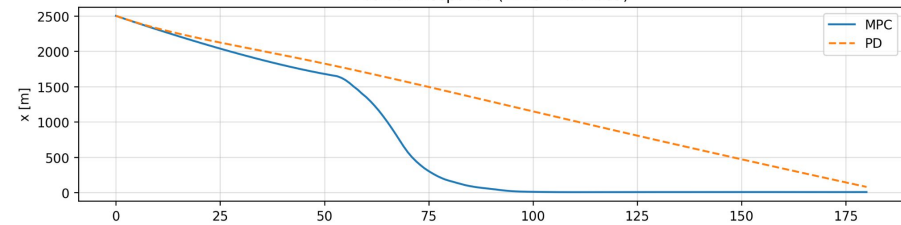
3D Landing Trajectory (Wind-Disturbed)



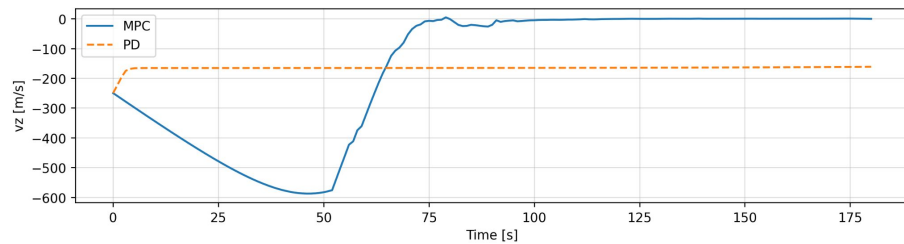
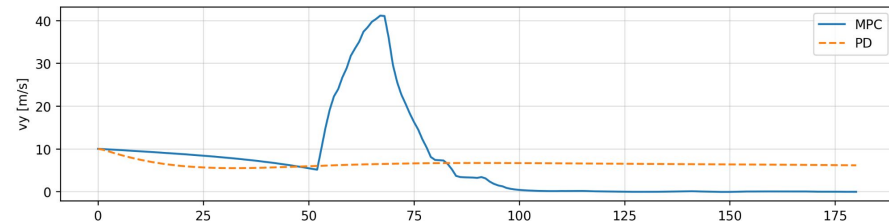
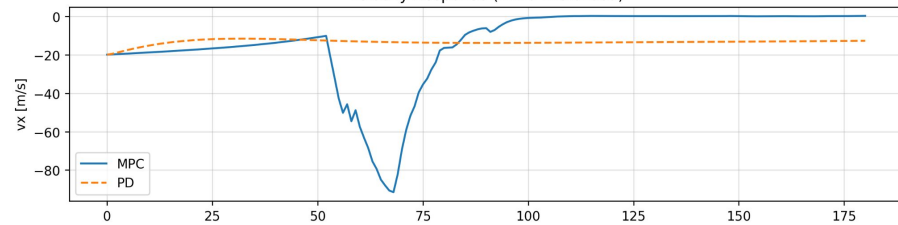
Robustness Study - Wind disturbances

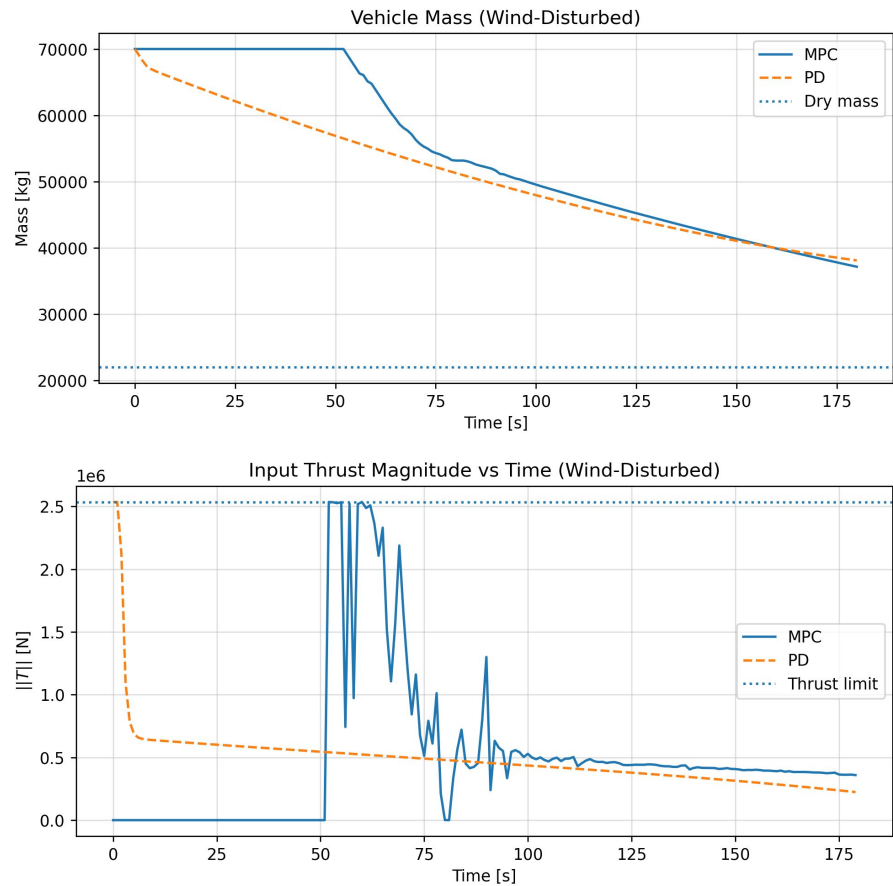
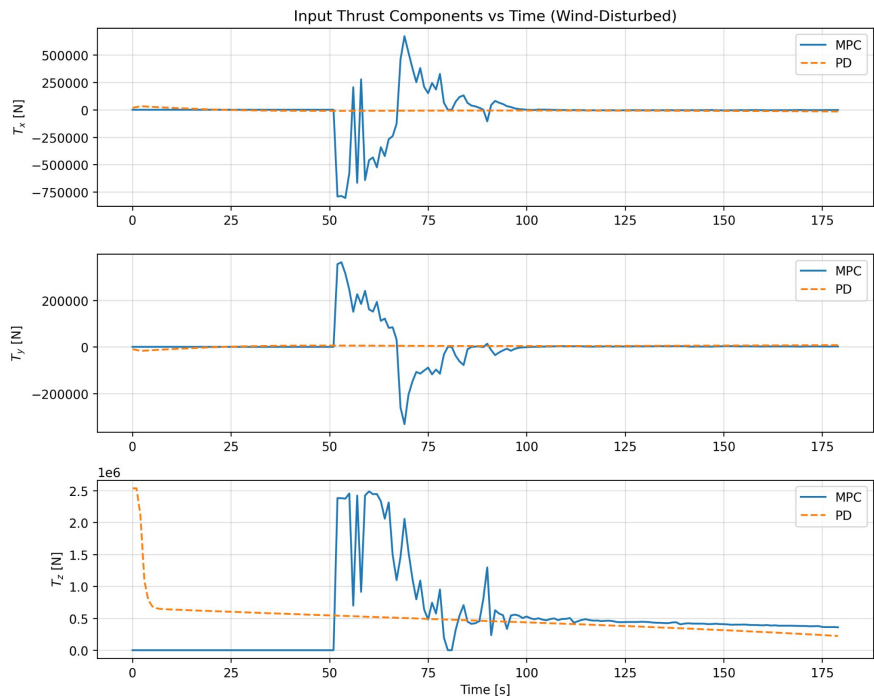
	MPC	PD
Final Position Error [m]	8.294	162.144
Final Horizontal Error [m]	7.988	88.485
Final Altitude [m]	2.233	135.872
Final Speed [m/s]	0.502	162.089
Fuel Used [kg]	32850.517	31896.447
Fuel Remaining [kg]	15149.483	16103.553
Max Thrust [N]	2535000.000	2535000.000
Max Tilt Angle [deg]	20.000	4.372

Position Response (Wind-Disturbed)

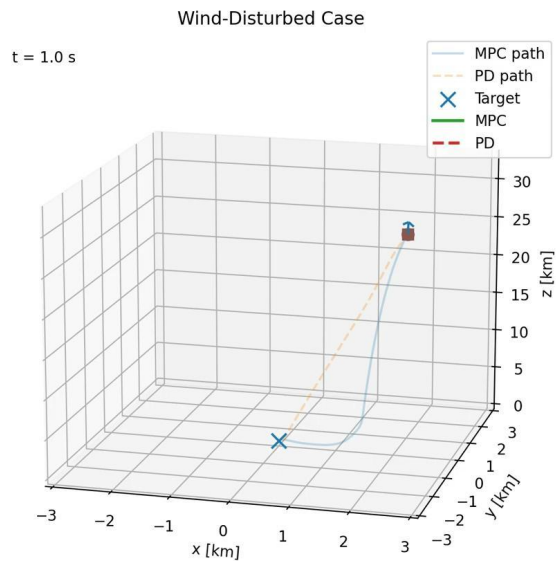


Velocity Response (Wind-Disturbed)

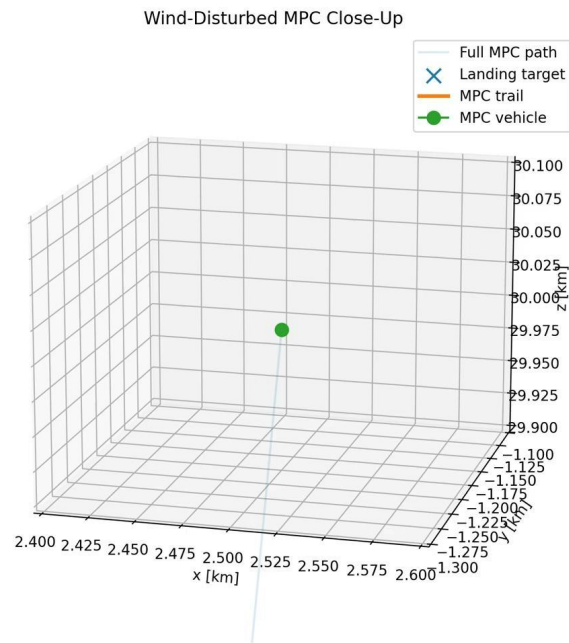




Wind Animation(s)



t = 0.0 s
x = 2500.0 m
y = -1200.0 m
z = 30000.0 m
speed = 251.00 m/s
|T| = 0.0 N



MPC Comparison

MPC	Nominal	Wind	Change
Final Position Error [m]	1.453	8.294	+6.841
Final Horizontal Error [m]	0.004	7.988	+7.984
Final Altitude [m]	1.453	2.233	+0.780
Final Speed [m/s]	0.122	0.502	+0.380
Fuel Used [kg]	26,124.075	32,850.517	+6,726.442
Fuel Remaining [kg]	21,875.925	15,149.483	-6,726.442
Max Thrust [N]	2,532,735.227	2,535,000.000	+2,264.773
Max Tilt Angle [deg]	20.000	20.000	N/A

Computation and Limitations

MPC	Nominal	Wind
Solves/iterations	134	180
Mean Solve time [s]	3.18	5.16
Max solve time [s]	16.52	23.93
Tolerance success rate [%]	82.1	88.3

Total of 180 seconds allowed for control
20 s horizon

50 optimization iterations at Δt of 1 s
Optimization tolerance of $1e-4$

- Wind is modeled as a constant lateral acceleration
 - An oversimplification
- Current MPC model is not ready for implementation into an actual real-time flight situation
 - Booster is a 3D point mass
 - No attitude dynamics
 - Drag model is simplified
- The model was also inconsistent in it's solving
 - This 134 sec nominal solve was a good run
 - Often, the nominal case would converge at ~155 sec.'

Conclusion

- A constrained finite-horizon MPC controller was implemented for a simplified 3D rocket booster landing problem
- The MPC was compared against a PD controller
- In the ideal case, the MPC successfully achieved engine cutoff
 - Final position error of 1.453m, and speed of 0.122 m/s
- The PD controller did not achieve the landing condition in either ideal or disturbed cases
- Under wind disturbances, the MPC did not achieve engine cutoff due to the horizontal error
 - Final position error of 8.294m, final speed of 0.502 m/s

Overall, the MPC was able to achieve better results as it could plan/predict a trajectory, handle the given constraints, and reduce the error more effectively than PD control.